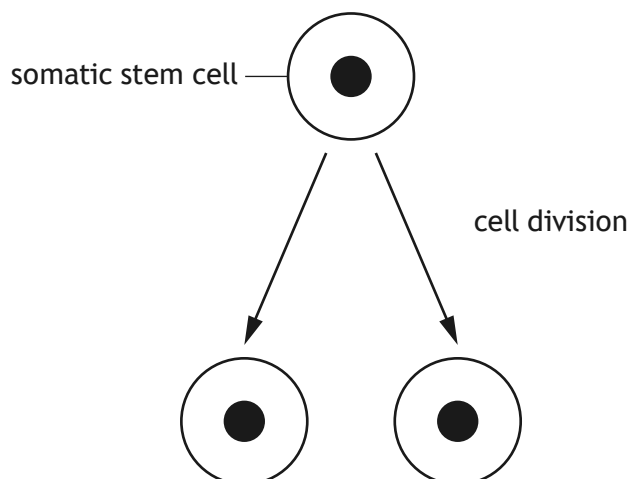


Total marks — 95

Attempt ALL questions

Question 15 contains a choice

1. The diagram shows a somatic stem cell dividing.



- (a) (i) Name the type of cell division shown in the diagram.

1

- (ii) Germline stem cells undergo a different type of cell division to produce gametes.

State one location of the production of gametes.

1

- (b) Somatic stem cells can be found in the cornea of the eye.

Describe how these stem cells could differentiate into specialised cells such as corneal cells.

1



1. (continued)

- (c) Some conditions can cause severe damage to the cornea and somatic stem cells found in the eye. Stem cell transplants can be carried out to replace the damaged somatic stem cells with healthy stem cells.

The table shows the number of stem cell donors and the number of stem cell transplant recipients in one year.

Age group (years)	Number of stem cell donors	Number of stem cell transplant recipients
0–17	5	30
18–34	24	290
35–49	88	330
50–59	192	373
60–69	384	590
70–79	700	1021
80–99	394	656

- (i) Use data from the table to describe the changes that occur in the number of stem cell donors as age increases from 0 to 99 years.

2

- (ii) Using information in the table, suggest a benefit of using a patient's own stem cells for this treatment.

1

- (d) Corneal repair is a therapeutic use of stem cells.
Describe one use of stem cells in research.

1

